Lecture Scribe Example

L21_L22.pdf
PDF

I want a lecture scribe to refer for an exam. Here attached is the lecture notes for which I want you to give me exam ready scribe. The scribe should function as exam reference material and include proofs, derivations, and worked examples exactly as presented in the lecture.

Lecture Scribe Course:
Lecture number:
Topic:
Date:

Definitions:
(Key definitions)

Theorems / Propositions:
(Statements)

Proofs:
(Main proof ideas / steps)

Derivations / Equations:
(Important derivations)

Examples:
Given:
Find:
Solution:
Remarks:
Assumptions:
Special cases:
Summary:
Key results:

Above written is an example Scribe. Using the same

structure, generate a lecture scribe for the provided context.

Here is your **exam-ready lecture scribe** strictly structured from the provided lecture notes (L21–L22: Inequalities and Tail Bounds)

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lecture Number: 21 & 22
Topic: Inequalities and Tail Bounds
Date: March 24 & 26, 2025

Definitions

1. Tail of a Random Variable

The **tail** of a random variable X is:

$$P(X > x)$$

→ Represents probability of **rare/extreme events** (large deviations).

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2. Tail Event

An event of the form:

$$P(X \geq k)$$

→ Represents **large deviation from mean / overload events**

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3. Binomial Distribution Tail

If $X \sim \text{Binomial}(n, p)$:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

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4. Poisson Distribution Tail

If $X \sim \text{Poisson}(\lambda)$:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Theorems / Propositions

1. Markov's Inequality

For non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

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2. Chebyshev's Inequality

$$P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

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3. Chernoff Bound (Upper Tail)

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

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4. Chernoff Bound (Lower Tail)

$$P(X \leq a) \leq \min_{t<0} \frac{E[e^{tX}]}{e^{ta}}$$

Proofs

1. Proof of Markov's Inequality (Idea)

Start from expectation:

$$E[X] = \sum_{x} x \cdot p_X(x)$$

Consider only values where $X \geq a$:

$$E[X] \geq \sum_{x \geq a} x \cdot p_X(x)$$

Replace $x \geq a$ with a:

$$E[X] \geq \sum_{x \geq a} a \cdot p_X(x)$$

$$E[X] \geq a \cdot P(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

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2. Proof of Chebyshev's Inequality

Apply Markov on squared deviation:

$$P(|X - \mu| \geq a) = P((X - \mu)^2 \geq a^2)$$

Using Markov:

$$\leq \frac{E[(X - \mu)^2]}{a^2}$$

$$= \frac{\mathrm{Var}(X)}{a^2}$$

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3. Proof of Chernoff Bound (Core Idea)

Step 1: Transform event

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Step 3: Optimize over t

$$P(X \geq a) \leq \min_{t > 0} \frac{E[e^{tX}]}{e^{ta}}$$

Derivations / Equations

1. Why Tail Bounds?

Exact computation:

$$P(X \geq k) = \sum(\text{long sums})$$

→ Computationally expensive
→ Use bounds:

$$P(X \geq k) \leq \text{simple expression}$$

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2. Chernoff Bound Key Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

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3. Moment Generating Function (Poisson Case)

For $X \sim \text{Poisson}(\lambda)$:

$$E[e^{tX}] = e^{\lambda(e^t - 1)}$$

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4. Chebyshev on Binomial Example

$$X \sim \text{Binomial}(n, \tfrac{1}{2}), \quad \mu = \tfrac{n}{2}, \quad \text{Var}(X) = \tfrac{n}{4}$$

$$P\left(X \geq \tfrac{3n}{4}\right) \leq \frac{1}{n}$$

Examples

Example 1: Binomial Tail (Servers)

Given:

Jobs distributed randomly across servers
$X \sim \text{Binomial}(n, p)$

Find:

$$P(X \geq k)$$

Solution:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Remarks:

- Rare event: one server overloaded
- Hard to compute exactly

Example 2: Poisson Tail (Arrivals)

Given:
$X \sim \text{Poisson}(\lambda)$

Find:

$$P(X \geq k)$$

Solution:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Remarks:

- Represents burst arrivals
- Tail = sudden spikes

Example 3: Markov Inequality (Coin Toss)

Given:
$X \sim \text{Binomial}(n, \frac{1}{2}), \mu = \frac{n}{2}$

Find:

$$P(X \geq \tfrac{3n}{4})$$

Solution:

$$P(X \geq \tfrac{3n}{4}) \leq \frac{E[X]}{3n/4} = \frac{n/2}{3n/4} = \frac{2}{3}$$

Remarks:

- Weak bound (independent of n)

Example 4: Income Distribution (Markov)

Given:

Mean income = 10,000

Find:

$$P(X \geq 50,000)$$

Solution:

$$\leq \frac{10000}{50000} = 0.2$$

Result:

At most 20% earn ≥ 50,000

Example 5: Exam Grades (Markov)

Given:

Mean = 70, threshold = 90

Find:

$$P(X \geq 90)$$

Solution:

$$\leq \frac{70}{90} = \frac{7}{9} \approx 0.778$$

Example 6: Server Load (Markov)

Given:

Mean = 5000, threshold = 20000

Find:

$$P(X \geq 20000)$$

Solution:

$$\leq \frac{5000}{20000} = \frac{1}{4}$$

Example 7: Chebyshev (Exam Scores)

Given:
Mean = 70, std = 10

Find:

$$P(|X - 70| \geq 30)$$

Solution:

$$\leq \frac{10^2}{30^2} = \frac{1}{9}$$

Example 8: Chebyshev (Grades)

Given:
$\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$

Find:

$$P(X \geq 90)$$

Solution:

$$P(|X - 70| \geq 20) \leq \frac{25}{400} = \frac{1}{16}$$

Result:

$$P(X \geq 90) \leq 0.0625$$

Example 9: Chebyshev (Network Latency)

Given:
$\mu = 50, \sigma^2 = 16$

Find:

$$P(|X - 50| \geq 20)$$

Solution:

$$\leq \frac{16}{400} = \frac{1}{25}$$

Result:
≤ 0.04

Remarks

- Tail probabilities = **rare but important events**
- Exact computation = **complex sums**
- Bounds give:
  - Simplicity
  - Guarantees
- Progression:
  - Markov → weak
  - Chebyshev → better
  - Chernoff → strongest (exponential decay)

Assumptions

- Markov: $X \geq 0$
- Chebyshev: finite mean and variance
- Chernoff:
  - Exponential moments exist
  - Independence (in many applications)

Special Cases

- Markov uses only mean → weakest
- Chebyshev uses variance → tighter
- Chernoff uses exponential → **tightest**

Summary

- Tail = probability of extreme deviation
- Direct computation = infeasible
- Inequalities provide upper bounds
- Increasing strength:

$$\text{Markov} < \text{Chebyshev} < \text{Chernoff}$$

Key Results

- $P(X \geq a) \leq \frac{E[X]}{a}$
- $P(|X - \mu| \geq a) \leq \frac{\text{Var}(X)}{a^2}$
- $P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$
- Chernoff gives **exponential decay bounds**
- Best bound achieved by optimizing t

If you want, I can convert this into a **1-page cheat sheet**, **flashcards**, or **exam questions with solutions**.